7116051

Quiz: Retrieval-Augmented Generation (RAG)

Name: _____ARYAN_____
Date: _____16/03/2026_____
Score: _____ / 20

Section A: Multiple Choice Questions (1 mark each)

1. What is the main purpose of Retrieval-Augmented Generation (RAG)?
A. To compress large datasets
B. To combine document retrieval with language generation
C. To train a neural network from scratch
D. To replace vector databases

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2. In a RAG system, what is the role of a vector database?
A. Store trained models
B. Store embeddings for similarity search
C. Generate answers
D. Train transformers

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3. What does “chunking” refer to in RAG pipelines?
A. Dividing a document into smaller sections
B. Encrypting documents
C. Compressing text data
D. Converting text into images

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4. Which method is commonly used to convert text chunks into vectors?
A. Tokenization

B. Embeddings
C. Sorting
D. Indexing

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5. Which component generates the final response in a RAG system?
A. Vector database
B. Embedding model
C. Large Language Model
D. Data loader

Section B: True or False (1 mark each)

6. RAG helps LLMs use external knowledge sources.
True / False

7. Chunking large documents improves retrieval accuracy.
True / False

8. Context injection means adding retrieved information into the prompt.
True / False

9. In RAG, the LLM directly searches PDFs without embeddings.
True / False

10. Evaluating a RAG system only requires checking model accuracy.
True / False

Section C: Short Answer (2 marks each)

11. Write the basic workflow of a RAG system.

Workflow: User query → Retrieve relevant chunks via embeddings from vector database → Inject retrieved context into LLM prompt → LLM generates final response.

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12. Why is chunk size important in RAG pipelines?

Chunk size matters because too large chunks may include irrelevant info, while too small chunks may lose context; optimal size improves retrieval accuracy.

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13. What is context injection and why is it useful?

Context injection means adding retrieved text into the LLM prompt; it ensures the model generates answers grounded in relevant external knowledge.

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14. Name two evaluation metrics or considerations for RAG systems.

Evaluation metrics: Retrieval relevance, answer accuracy (also faithfulness and latency can be considered).

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15. Give one real-world application of RAG.

Real-world application: Customer support chatbots that answer queries using product manuals and FAQs.

✅ **Total Marks: 20**